

Notation:

$m :=$ Menge aller Samples

$n :=$ Input feature / Eigenschaft

$L :=$ Anzahl der Layer

$l :=$ Layer Index

$w^l :=$ Matrix der Gewichte eines Layers.

$x_i :=$ Single Input Variable (

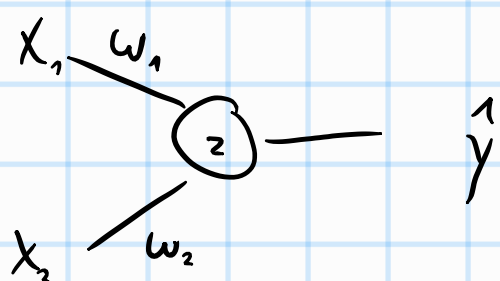
w^l
 ~~w^l~~
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Zusammenfassung

δ_j^i := Error of a Node

Einfach ein bisschen rumspielen

Forward pass:



Data
 $x_1 = 2$ $y = 0$
 $x_2 = 3$

$$z = \text{ReLU}((x_1 \cdot w_1) + (x_2 \cdot w_2) + b)$$

$$x_1 = 2 \quad x_2 = 3 \quad w_1 = 0,5 \quad w_2 = -1$$

$$z = \dots ((2 \cdot 0,5) + (3 \cdot (-1)) + 3)$$

$$z = \dots (-2 + 3)$$

$$z = \text{ReLU}(1)$$

$$z = 1$$

$$\Rightarrow \hat{y} = 1$$

Lernen:

$$L = y - \hat{y}$$

$$l = 0 - 1$$

$$l = -1$$

$$\alpha = 0,1 \quad \begin{array}{l} \text{Änderung} = \alpha \cdot l \cdot x_1 \\ \text{Änderung} := \Delta w_1 \end{array}$$

$$\Delta w_1 = 0,1 \cdot (-1) \cdot 2$$

$$\Delta w_1 = -0,2$$

$$w_{1, \text{neu}} = w_1 + \Delta w_1$$

$$w_{1, \text{neu}} = 0,3$$

$$\Delta w_2 = -1 \cdot 0,1 \cdot 3$$

$$\Delta b = 1 \cdot (-1) \cdot 0,1$$

$$\Delta w_2 = -0,3$$

$$\Delta b = -0,1$$

$$w_{2, \text{neu}} = -1 - 0,3$$

$$b_{\text{neu}} = 2,9$$

$$w_{2, \text{neu}} = -1,3$$

$$\text{Optimiert} \Rightarrow z_{\text{neu}} = 0,3 x_1 - 1,3 x_2 + 2,9$$

$$x_1 = 2 \quad x_2 = 3$$

$$z(2,3) = 0,3 \cdot 2 - 1,3 \cdot 3 + 2,9$$

$$z(2,3) = \text{ReLU}(-0,1)$$

$$z(2,3) = 0$$

Complete Calc. of one Network

Notation:

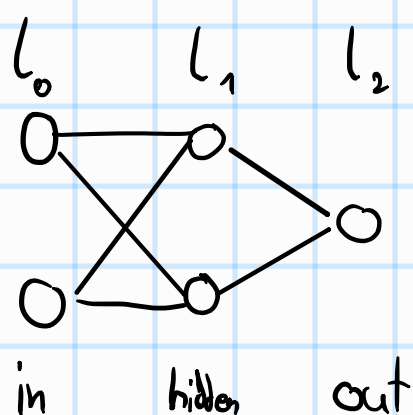
$l :=$ Schicht

$z^{(l)} :=$ Pre-Activation

$a^{(l)} :=$ Activation

$w_{idx, inp}^{(l)} :=$ Gewicht mit Stelle (idx) und Input (inp)

Architektur:



Input: $x_1 = 2$ $x_2 = 1$ $X = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

gewichte: $w_{1,1}^{(1)} = 0,5$ $w_{1,2}^{(1)} = -1$ $b_1^{(1)} = 0,5$

$w_{2,1}^{(1)} = -0,5$ $w_{2,2}^{(1)} = 0,5$ $b_2^{(1)} = -0,5$

-0,2

Forward propagation:

Berechnung von $l=1$

$$z_1^{(1)} = (w_{1,1}^{(1)} \cdot x_1) + (w_{1,2}^{(1)} \cdot x_2) + b_1^{(1)}$$

$$z_1^{(1)} = (0,5 \cdot x_1) + (-1 \cdot x_2) + 0,5$$

$$z_1^{(1)} = (0,5 \cdot 2) + (-1 \cdot 1) + 0,5$$

$$z_1^{(1)} = 0,5$$

$$a_1^{(1)} = \text{ReLU}(z_1^{(1)})$$

$$\underline{a_1^{(1)} = 0,5}$$

$$z_2^{(1)} = (w_{2,1}^{(1)} \cdot x_1) + (w_{2,2}^{(1)} \cdot x_2) + b_2^{(1)}$$

$$z_2^{(1)} = (-0,5 \cdot 2) + (0,5 \cdot 1) - 0,5$$

$$z_2^{(1)} = -1$$

$$a_2^{(1)} = \text{ReLU}(-1)$$

$$\underline{a_2^{(1)} = 0}$$

Berechnung von $L=2$ (Output Layer)

$$w_{1,1}^{(2)} = 1,0 \quad w_{1,2}^{(2)} = 2,0 \quad b_1^{(2)} = -0,2 \quad a_1^{(1)} = 0,5 \quad a_2^{(1)} = 0$$

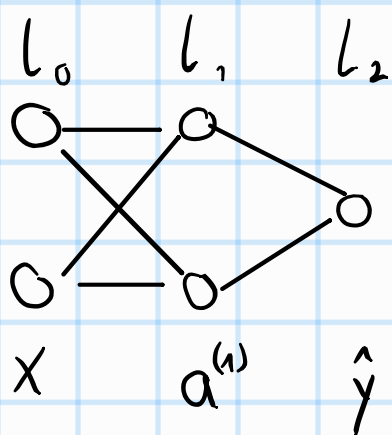
$$z_1^{(2)} = (w_{1,1}^{(2)} \cdot a_1^{(1)}) + (w_{1,2}^{(2)} \cdot a_2^{(1)}) + b_1^{(2)}$$

$$z_1^{(2)} = (1,0 \cdot 0,5) + (2,0 \cdot 0) - 0,2$$

$$z_1^{(2)} = 0,5 - 0,2$$

$$z_1^{(2)} = 0,3$$

$$a_1^{(2)} = 0,3$$

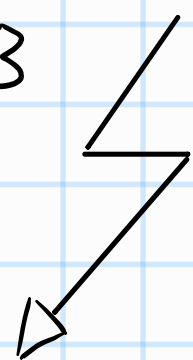


$$\text{Input: } X : \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\text{Hidden Layer Activations } a^{(1)} : [0,5; 0]$$

$$\text{prediction } \hat{y} : 0,3$$

$$\text{Label } y : 1,0$$



Backpropagation:

1. Loss bestimmen:

praktisch für Ableitung

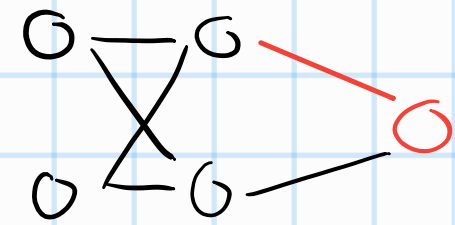
$$\begin{aligned} E(y, \hat{y}) &= \frac{1}{2} (y - \hat{y})^2 \\ &= \frac{1}{2} (1 - 0,3)^2 \\ &= \frac{1}{2} \cdot 0,49 = 0,245 \end{aligned}$$

2.1 Gradientenbestimmung ($w_{1,1}^{(2)}$ $E(a_1^{(2)}(z_1^{(2)}(\dots)))$)

Loss Glied Aktivierungsglied Gewichtsglied

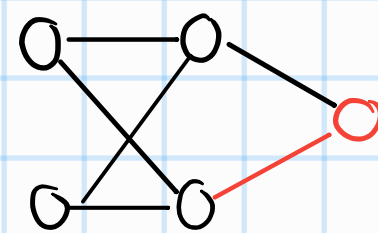
$$\frac{\partial E}{\partial w_{1,2}^{(2)}} = \frac{\partial E}{\partial \hat{y}} \cdot \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \cdot \frac{\partial z_1^{(2)}}{\partial w_{1,1}^{(2)}}$$

$z_1^{(2)}(w_{1,1}^{(2)}, w_{1,2}^{(2)}) = (w_{1,1}^{(2)} \cdot a_1^{(1)}) + (w_{1,2}^{(2)} \cdot a_2^{(1)}) + b_1^{(2)}$



$$\begin{aligned} \frac{\partial E}{\partial \hat{y}} &= -y + \hat{y} = -1 + 0,3 = -0,7 \\ \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} &= 1 \cdot z_1^{(2)} = 1 \\ \frac{\partial z_1^{(2)}}{\partial w_{1,1}^{(2)}} &= a_1^{(1)} = 0,5 \end{aligned}$$
$$\frac{\partial E}{\partial w_{1,2}^{(2)}} = -0,7 \cdot 1 \cdot 0,5 = -0,35$$

2.2 Gradientenbestimmung ($w_{1,2}^{(2)}$) $E(a_1^{(2)}(z_1^{(2)}(\dots)))$



$$\frac{\partial E}{\partial w_{1,2}^{(2)}} = \frac{\partial E}{\partial a_1^{(2)}} \cdot \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \cdot \frac{\partial z_1^{(2)}}{\partial w_{1,2}^{(2)}}$$

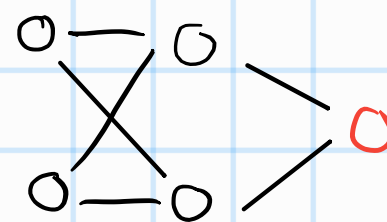
$$z_1^{(2)} = (w_{1,1}^{(2)} \cdot a_1^{(1)}) + (w_{1,2}^{(2)} \cdot a_2^{(1)} + b$$

<u>Loss Glied</u>	<u>Aktivierung Glied</u>	<u>Gewichts Glied</u>
$\frac{\partial E}{\partial a_1^{(2)}} = -0,7$	$\frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} = 1$	$\frac{\partial z_1^{(2)}}{\partial w_{1,2}^{(2)}} = a_2^{(1)} = 0$

$$\frac{\partial E}{\partial w_{1,2}^{(2)}} = -0,7 \cdot 1 \cdot 0$$

$$= 0$$

2.3 Gradientenbestimmung ($b_1^{(2)}$) $E(a_1^{(2)}(z_1^{(2)}(\dots)))$

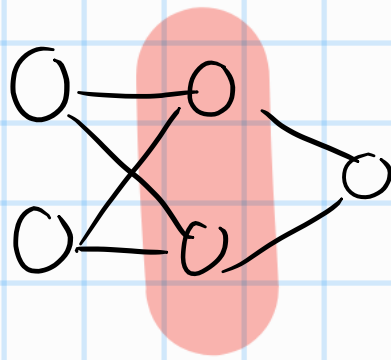


$$\frac{\partial E}{\partial b_1^{(2)}} = \frac{\partial E}{\partial a_1^{(2)}} \cdot \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \cdot \frac{\partial z_1^{(2)}}{\partial b_1^{(2)}}$$

$$\frac{\partial E}{\partial a_1^{(2)}} = -0,7 \quad \left| \quad \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} = 1 \quad \right| \quad \frac{\partial z_1^{(2)}}{\partial b_1^{(2)}} = 1$$

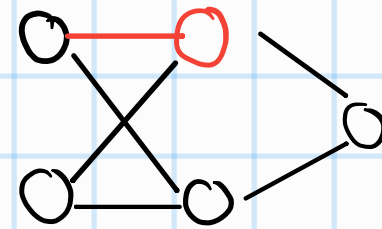
$$\frac{\partial E}{\partial b_1^{(2)}} = -0,7 \cdot 1 \cdot 1 = -0,7$$

3.0 Gradientenbestimmung $L = 1$



3.1 Gradientenbestimmung von $(w_{1,1}^{(1)})$ $E(a_1^{(2)}(z_1^{(2)}(a_1^{(1)}(z_1^{(1)}(\dots)), a_2^{(1)}(\dots))))$

$$\frac{\partial E}{\partial w_{1,1}^{(1)}} = \frac{\partial E}{\partial a_1^{(2)}} \cdot \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \cdot \underbrace{\frac{\partial z_1^{(2)}}{\partial a_1^{(1)}} \cdot \frac{\partial a_1^{(1)}}{\partial z_1^{(1)}} \cdot \frac{\partial z_1^{(1)}}{\partial w_{1,1}^{(1)}}}_{\text{neuer Teil}}$$



$$z_1^{(1)} = (w_{1,1}^{(1)} \cdot x_1) + (w_{1,2}^{(1)} \cdot x_2)$$

$$a_1^{(1)} = \text{ReLU}(z_1^{(1)})$$

$$z_1^{(2)} = (w_{1,1}^{(2)} \cdot a_1^{(1)}) + (w_{1,2}^{(2)} \cdot a_2^{(1)})$$

$$\text{ReLU}(x) = \begin{cases} 0 & \text{für } x \leq 0 \\ x & \text{für } x > 0 \end{cases}$$

$$\frac{\partial E}{\partial a_1^{(2)}} = -0,7$$

$$\frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} = 1$$

$$\frac{\partial z_1^{(2)}}{\partial a_1^{(1)}} = w_{1,1}^{(2)} = 1,0$$

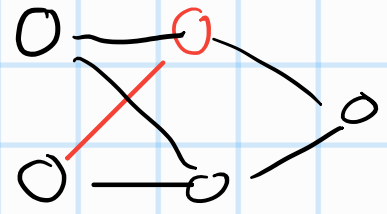
$$\frac{\partial a_1^{(1)}}{\partial z_1^{(1)}} = 1$$

$$\frac{\partial z_1^{(1)}}{\partial w_{1,1}^{(1)}} = 2$$

$$\frac{\partial E}{\partial w_{1,1}^{(1)}} = -0,7 \cdot 1 \cdot 1 \cdot 1 \cdot 2 = -1,4$$

3.2 Gradientenbestimmung von $(w_{1,2}^{(1)}) \in (a_1^{(2)}(z_1^{(2)}(a_1^{(1)}(z_1^{(1)}(w))))$

$$\frac{\partial E}{\partial w_{1,2}^{(1)}} = \frac{\partial E}{\partial a_1^{(2)}} \cdot \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \cdot \frac{\partial z_1^{(2)}}{\partial a_1^{(1)}} \cdot \frac{\partial a_1^{(1)}}{\partial z_1^{(1)}} \cdot \frac{\partial z_1^{(1)}}{\partial w_{1,2}^{(1)}}$$



$$z_1^{(2)} = (w_{1,1}^{(2)} \cdot a_1^{(1)}) + (w_{1,2}^{(2)} \cdot \dots)$$

$$\frac{\partial E}{\partial a_1^{(2)}} = -0,7 \Rightarrow \left| \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} = 1 \right| \left| \frac{\partial z_1^{(2)}}{\partial a_1^{(1)}} = w_{1,1}^{(2)} = 1,0 \right|$$

$$\left| \begin{array}{l} a_1^{(1)} = \text{ReLU}(z_1^{(1)}) = 0,5 \\ \frac{\partial a_1^{(1)}}{\partial z_1^{(1)}} = 1 \end{array} \right| \quad \begin{array}{l} \frac{\partial z_1^{(1)}}{\partial w_{1,2}^{(1)}} = x_2 \\ = 1 \end{array}$$

$$-0,7 \cdot 1 \cdot 1 \cdot 1 \cdot 1$$